

Day 04 –TCP/IP-OSI

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THEORY

1. What is a network protocol?

A set of rules defining how data should be communicated between devices over a network.

2. What is the OSI Model?

Open system interconnection.

It was made to create an international ,vendor neutral network standards

3. Name all seven OSI layers in correct order.

- 7.application layer
- 6.presentation layer
- 5.session layer.
- 4.transport layer
- 3.network layer
- 2.data link layer
- 1.physical layer

4. What is the purpose of the Application Layer?

It defines how application processes send , receive and interpret data.

5. What is the purpose of the Transport Layer?

provides end to end communications between application processes also known as process to process or service to service communication.

6. What is the purpose of the Network Layer?

Providing logical addressing and routing (ip address).

7. What is the purpose of the Data Link Layer?

Local network communication (mac address)

8. What is the purpose of the Physical Layer?

Sends and receives bits as electrical , optical or radio signals over the medium (like cables).

9. What is the TCP/IP Model?

A way of grouping protocols into layers based on their functions.

It is mainly based on transfer control protocol(TCP) and internet protocol(IP).

10. Explain the difference between OSI and TCP/IP models.

The main difference is the layers

In osi model application + presentation +session = application in tcp/ip

In osi model transport = transport in tcp/ip

In osi model network layer = internet layer in tcp/ip

In osi model data link +physical layer = network access in tcp/ip

11. What is encapsulation?

It is moving down the network stack to prepare data for travelling on physical medium by adding headers or trailers(data---->segment/datagrams---->packet----->frame----->bits)

12. What is decapsulation?

It is moving up the network stack to transfer bits to data by removing heads and trailers

13. Which layer uses IP addresses?

Network layer

14. Which layer uses MAC addresses?

Data link layer

15. Which layer uses TCP and UDP?

Transport Layer.

16. Which layer includes HTTP and HTTPS?

Application layer.

17. Why should cybersecurity professionals understand networking layers?

To understand how data moves through a network, identify vulnerabilities, detect attacks, and troubleshoot security problems.

18. Explain how network data travels from a browser to a server.

The browser sends a request through the networking layers. The data is divided into packets, given addressing information such as IP and port numbers, and sent across the network. The server receives the packets, processes the request, and sends a response back through the same layers.

PRACTICAL

19. Your resolved IP address for google.com (nslookup).

ipv6

2a00:1450:4025:1807::65
2a00:1450:4025:1807::66
2a00:1450:4025:1807::64
2a00:1450:4025:1807::71

ipv4

172.217.23.174

20. Number of traceroute hops to google.com

10 hops

1	3 ms	4 ms	1 ms	dev.opt [192.168.100.1]
2	4 ms	3 ms	5 ms	41.34.4.1
3	8 ms	3 ms	6 ms	10.45.10.53
4	5 ms	4 ms	6 ms	10.35.14.2
5	5 ms	5 ms	4 ms	10.45.28.66
6	5 ms	11 ms	5 ms	TELECOM-EGY.edge1.Marseille3.Level3.net [213.19.202.134]

```
7  51 ms  53 ms  50 ms  et-0-0-28.edge1.Marseille3.Level3.net [213.19.201.81]
8  *      *      *      Request timed out.
9  40 ms  41 ms  39 ms  142.250.163.174
10 49 ms  46 ms  46 ms  mu-in-f102.1e100.net [192.178.194.102]
```

SCREENSHOTS

1.ping

```
PS C:\Windows\system32> ping google.com

Pinging google.com [142.251.209.46] with 32 bytes of data:
Reply from 142.251.209.46: bytes=32 time=48ms TTL=114
Reply from 142.251.209.46: bytes=32 time=54ms TTL=114
Reply from 142.251.209.46: bytes=32 time=64ms TTL=114
Reply from 142.251.209.46: bytes=32 time=53ms TTL=114

Ping statistics for 142.251.209.46:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 48ms, Maximum = 64ms, Average = 54ms
PS C:\Windows\system32> |
```

2.tracert

```
PS C:\Windows\system32> tracert google.com

Tracing route to google.com [192.178.194.102]
over a maximum of 30 hops:

  0  3 ms  4 ms  1 ms  dev.opt [192.168.100.1]
  1  4 ms  3 ms  5 ms  41.34.4.1
  2  8 ms  3 ms  6 ms  10.45.10.53
  3  5 ms  4 ms  6 ms  10.35.14.2
  4  5 ms  5 ms  4 ms  10.45.28.66
  5  5 ms  11 ms  5 ms  TELECOM-EGY.edge1.Marseille3.Level3.net [213.19.202.134]
  6  51 ms  53 ms  50 ms  et-0-0-28.edge1.Marseille3.Level3.net [213.19.201.81]
  7  *      *      *      Request timed out.
  8  40 ms  41 ms  39 ms  142.250.163.174
  9  49 ms  46 ms  46 ms  mu-in-f102.1e100.net [192.178.194.102]

Trace complete.
```

3.nslookup

```
PS C:\Windows\system32> nslookup google.com
```

```
Server: dev.opt
```

```
Address: fe80::1
```

```
Non-authoritative answer:
```

```
Name: google.com
```

```
Addresses: 2a00:1450:4025:1807::65
```

```
2a00:1450:4025:1807::66
```

```
2a00:1450:4025:1807::64
```

```
2a00:1450:4025:1807::71
```

```
172.217.23.174
```